SIMATS ENGINEERING
Slot: A, Course Code: ITA0513, Course Name: Computer Vision

Duration: 60 mins Marks: 20 Assessment Tool 2: Concept Mapping (Medium) Assessment Tool Weightage: 20%

COURSE OUTCOME COVERED

CO1: *Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems.* (BTL 3)

Q. No.	Questions
1	Develop an original and analytically structured concept map representing image formation models, identifying illumination, object properties, geometric projection and camera parameters, and connecting their interactions to image quality and computer vision applications.
2	Construct a logically organized concept map illustrating sampling and quantization, identifying their fundamental parameters, establishing relationships with spatial and intensity resolution, and analysing their combined influence on image quality and subsequent processing.

Code of Conduct:

*I **Midhuna A (192521302)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student
Midhuna A
192521302

RUBRICS FOR CALCULATION

Criteria	Weigh tage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure

Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

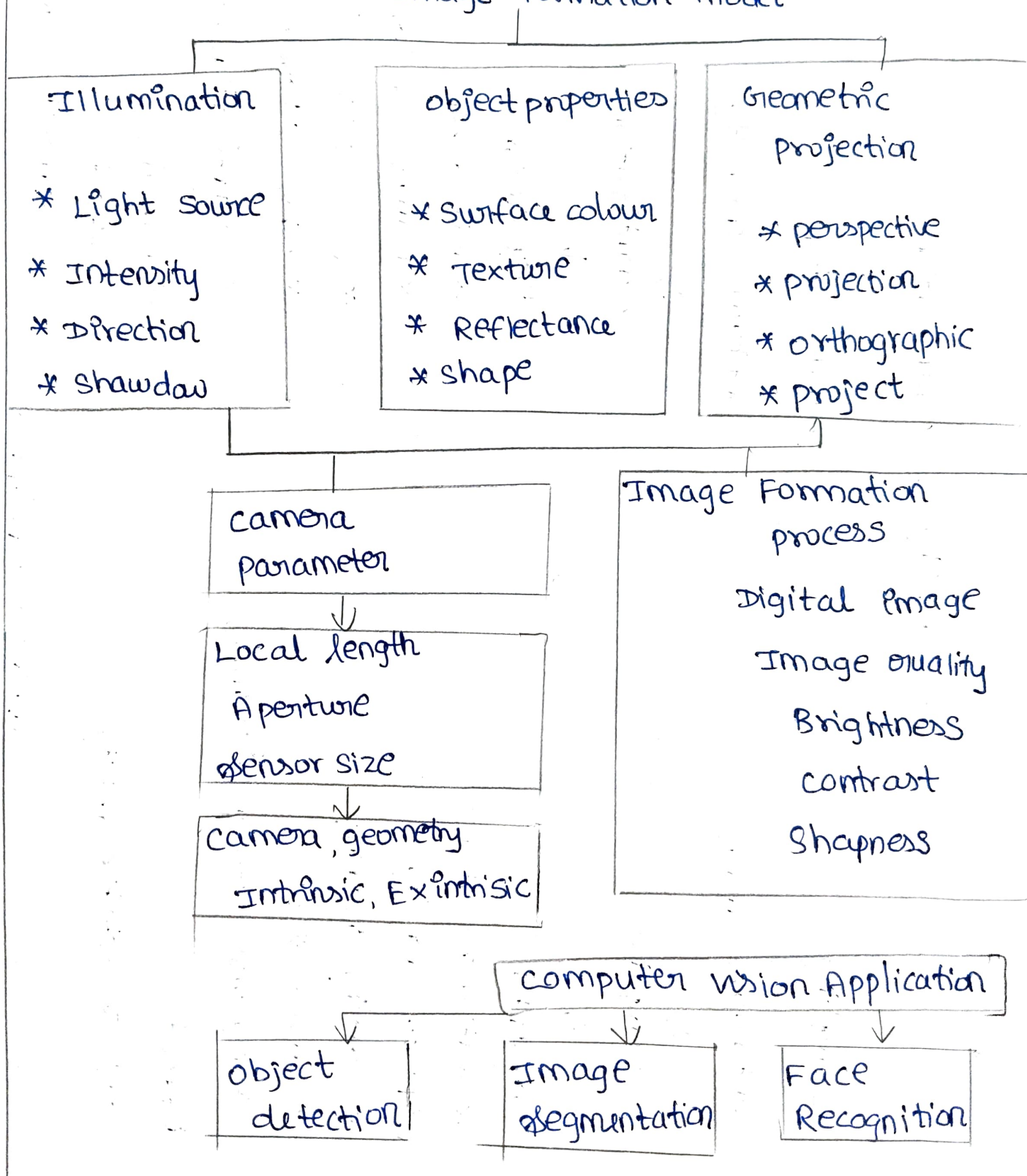
CO1 AT2 concept mapping

Name : A. Midhuna

Register No : 192521302

Date : 7/10/26

1.



2. Sampling & Quantization

Sampling

continuous Image

↓

Sampling

↓

Key factor :

* Spatial resolution
* pixcel density
* Sampling frequency
* Image Size
* Nyquist criterion (fs ≥ fmax)

Effect:

object detection ability

Quantization

continuous Intensity

↓

Quantization

↓

Key factor

* Bit depth
* number of level ($L = 2^n$)
* 8 bit = 256 level
* 2 bit = 4 levels
* quantization error
* Banding

effect :

smooth Gradients

Combined Influence

Spactial Resolution + Intensity Resolution